EXPERIMENT 12: NMAP TO DISCOVER LIVE HOSTS

AIM:

To discover live hosts using ARP, ICMP and TCP/UDP host-discovery scans with Nmap on an authorized lab network.

PROCEDURE:

1. Use a private lab network or the assigned TryHackMe room and ensure the target range is explicitly authorized.

2. Install Nmap if it is not already available.

3. Identify the assigned target IP address or subnet.

4. Run an ARP discovery scan on a local Ethernet subnet, or use ICMP/TCP discovery options as permitted.

5. Record which hosts are reported as up and compare scan methods.

6. Do not scan networks without authorization.

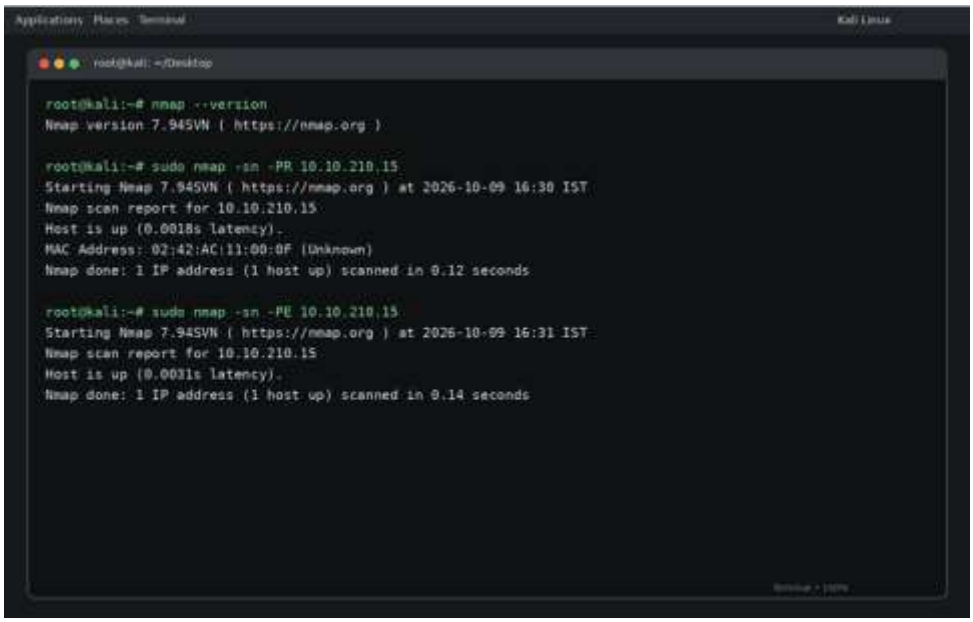
7. Check that the target range is the one assigned for the practical and is within the authorized lab environment.
8. Compare the hosts reported by each discovery method and note that firewalls may block some probes.
9. Record the scan options and the number of hosts reported as up.
10. Stop the scan if the target range is found to be outside the approved scope.

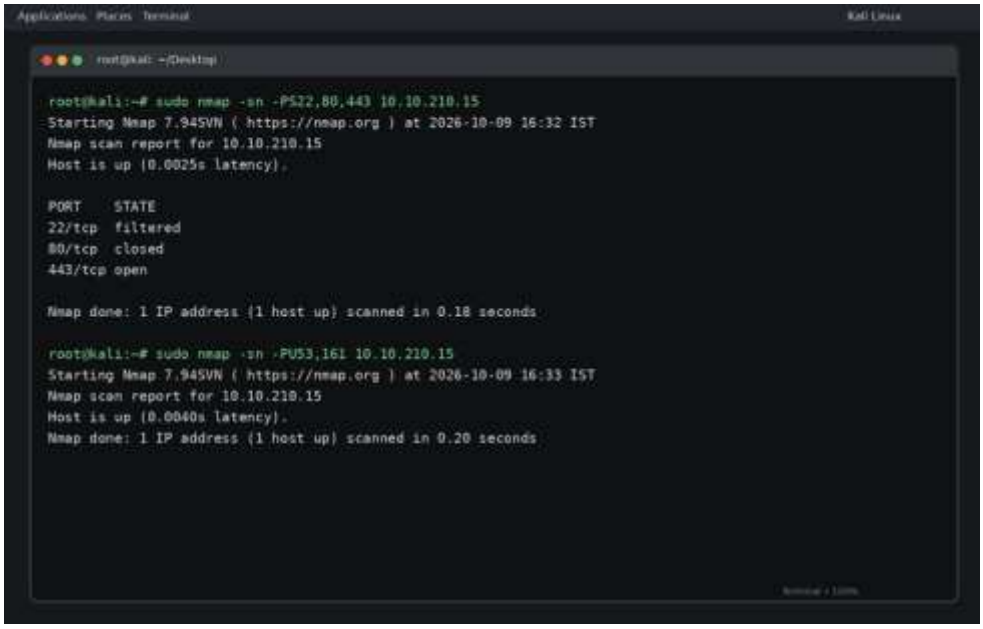
CODE:

```
# Replace <authorized-target> with the assigned lab host
or subnet.
nmap -sn -PR <authorized-target>
nmap -sn -PE <authorized-target>
nmap -sn -PS80,443 <authorized-target>
nmap -sn -PU53 <authorized-target>
```

```
# Example for a private lab subnet only:
nmap -sn -PE 192.168.56.0/24
```

OUTPUT:





RESULT:

Thus, ARP, ICMP and TCP/UDP-based Nmap host-discovery methods were studied on an authorized lab network.